

RiskPulse

An AI/NLP risk engine that turns news and tweets into structured financial risk signals

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88%

sentiment accuracy after fine-tuning
(from 72%)



89%

event-type accuracy

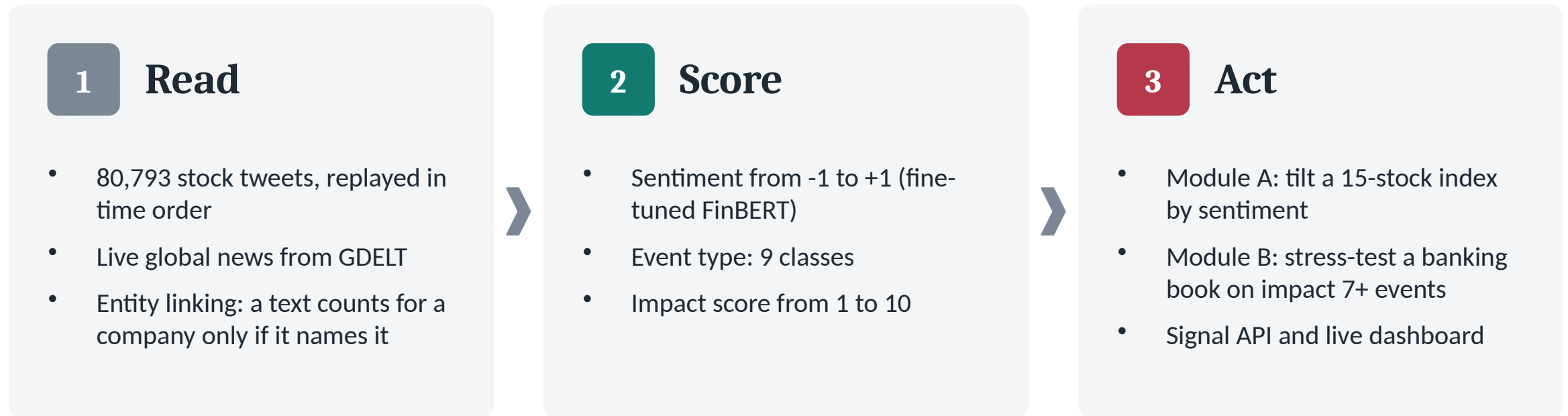


3.2x

normal price move on high-impact days

Risk appears in text before it appears in prices

A downgrade, a lawsuit, a Fed comment or a sanctions headline is written down before the market fully reacts. Risk teams cannot read thousands of posts an hour, so the early signal is missed.



System design: one engine, two consumers

RiskPulse: AI/NLP Risk Engine architecture

Unstructured text in, structured risk signals out, consumed by two downstream modules



Validation against real prices

Stock-days the engine scores 7-10 moved 3.2x their normal amount the same day (0.96x for scores 1-3), but only 1.2x the next day.

Module A edged equal weight by +0.14% a year after costs, too small to trade on: the engine is a risk monitor, not a trading signal. Module B: a +2% rate shock is the banking book's biggest risk (-10%).

Implementation choices, and why

Entity linking

A name, alias or cashtag must appear. It caught a data bug: 4,080 tweets copied under three companies.

Fine-tuned FinBERT

Fine-tuned on 8,588 finance tweets on a free Kaggle GPU. Held-out accuracy rose from 71.6% to 88.1%.

Event classifier

TF-IDF + logistic regression: trains in a minute and labels 57K texts in seconds on a CPU. Zero-shot would take hours.

Impact score

Event severity x confidence x sentiment strength x abnormal attention, using past days only (no look-ahead).

Signal API

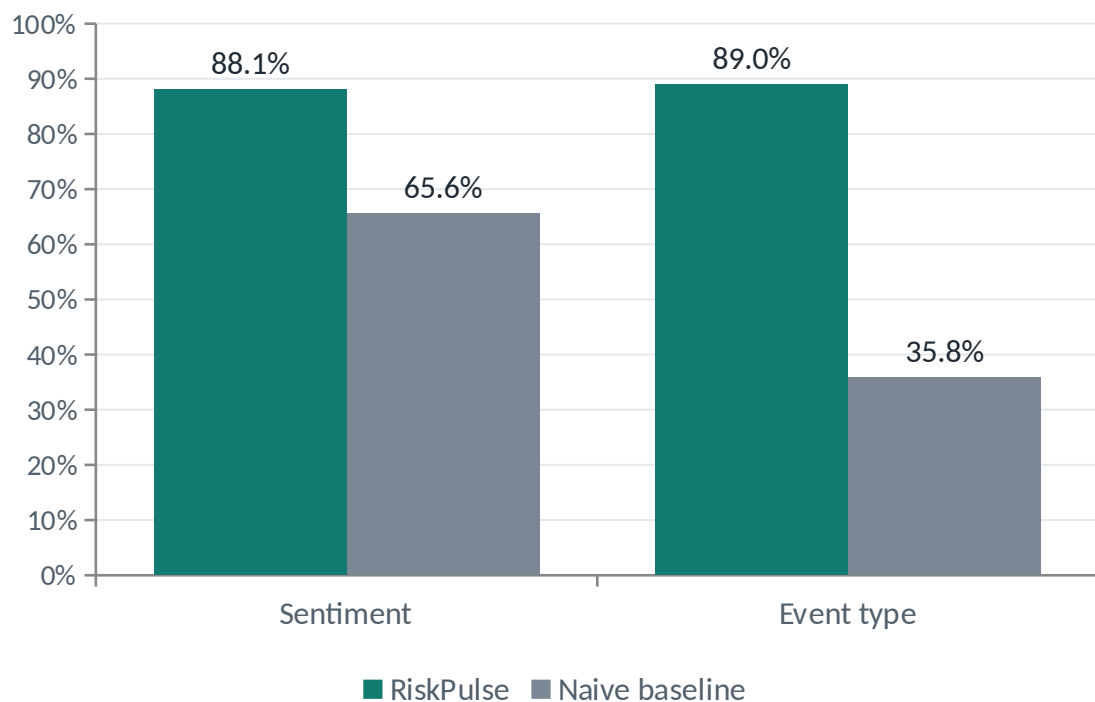
FastAPI endpoints: Module A subscribes to /daily, Module B to /signals/high-impact.

Honest backtests

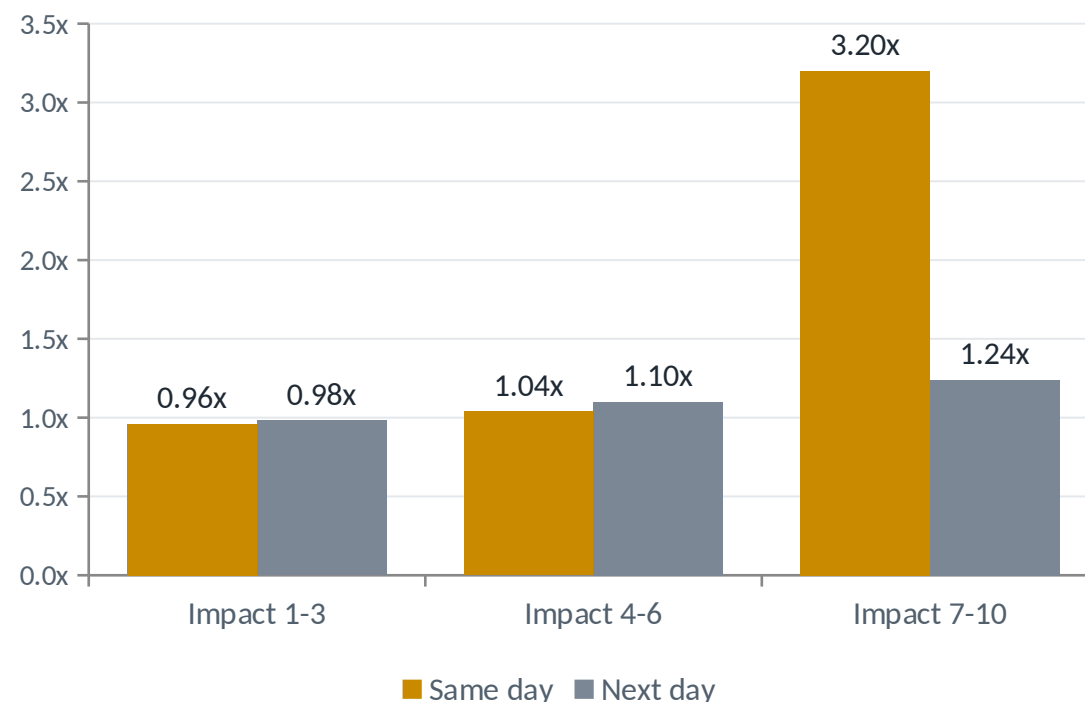
Equal-weight benchmark, 10 bps trading costs, and a 9-setting robustness grid.

Every component beats its naive baseline on unseen data

Accuracy on held-out labelled tweets



Price move vs the stock's normal day



Fine-tuning lifted sentiment from 71.6% (original FinBERT) to 88.1%. Macro-F1: sentiment 0.85 (original 0.66, naive 0.26), events 0.87. 56,931 signals; the impact score ranks price moves better than tweet volume alone (0.078 vs 0.059).

A real-time risk monitor, not a trading signal

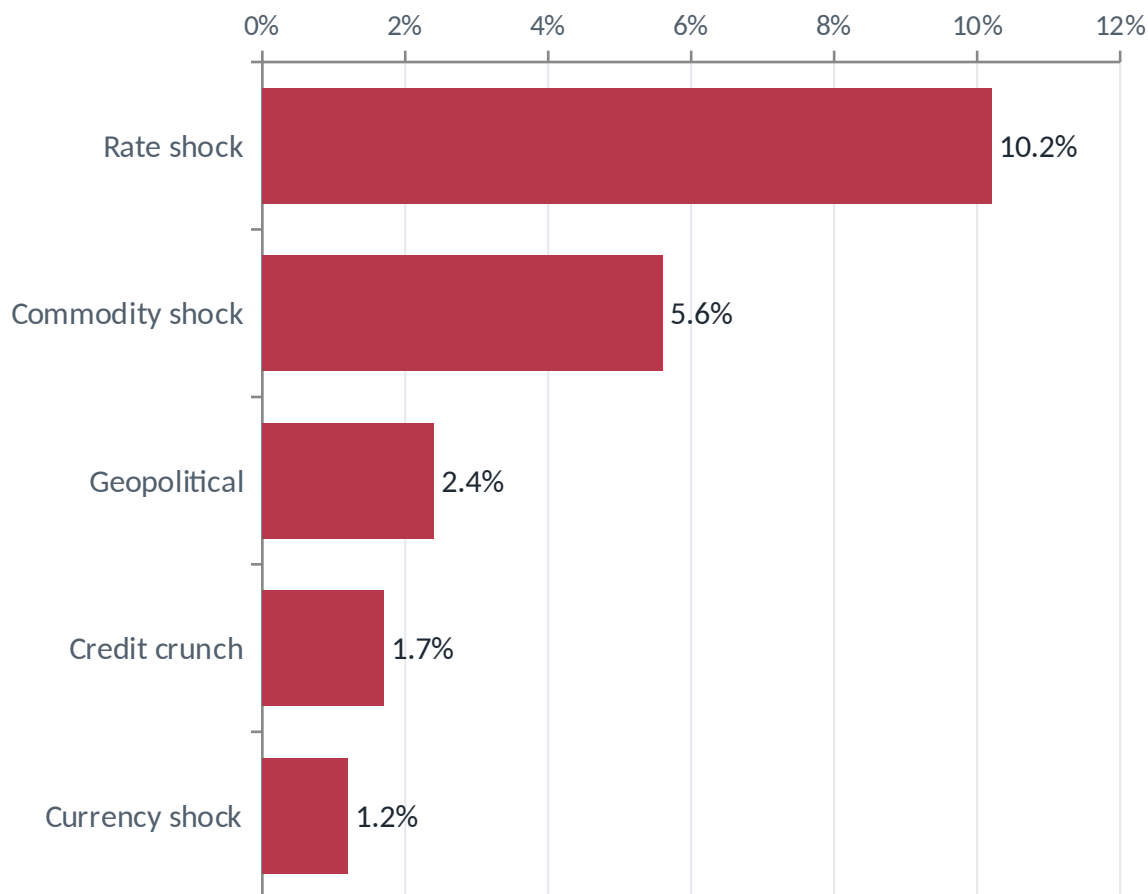
A Module A: sentiment-tilted index

After fine-tuning, it edged equal weight by +0.14% a year after 10 bps costs (6 of 9 settings positive), but that is too small to trade on: tweets mostly react to moves that already happened.

B Module B: event-driven stress tests

31 stress tests triggered on a synthetic \$862m banking book. Example: a story that Florida could face Disney's \$1B district debt (22 Apr 2022, impact 9) triggers a Disney credit shock costing \$14.2m.

Loss on the \$862m book (impact-8 event)



Limitations and next steps

Limitations

- Fine-tuned on news-style tweets; retail chatter ("EARNINGS THREAD") can still score strongly
- Word-based event model is fooled by slang ("Gold secured! \$TSLA")
- Event weights are hand-set; only 49 stock-days scored 7+
- Stress shocks are fixed per scenario, first-order valuation

Next steps

- Fine-tune further on labelled retail stock tweets
- Learn impact weights from realised price moves
- Size each shock to the specific event; full repricing
- Streaming news feed instead of rate-limited GDELT polling